

DEPI Graduation Project

Requirements Gathering

System Under Test: Swag Labs (SauceDemo)

<https://www.saucedemo.com>

Date: April 23, 2026

Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Role	Interest / Expectation
DEPI Graduate Student (Tester)	Project owner & QA engineer	Demonstrate professional testing skills; achieve graduation
DEPI Academic Supervisor	Evaluator	Adherence to documentation standards; test coverage quality
Sauce Labs (App Owner)	System provider	Demo app used correctly; no SLA obligations
End Users (Simulated)	Shoppers on Swag Labs	Seamless authentication, browsing, and checkout experience
Future Developers (Hypothetical)	Maintenance team	Regression suite enables safe code changes

3.2 User Stories & Use Cases

User Types (Provided by SauceDemo)

Username	Password	Behavior
standard_user	secret_sauce	Normal functioning user — baseline for all positive test cases
locked_out_user	secret_sauce	Receives error on login — tests locked account handling
problem_user	secret_sauce	Experiences broken images and incorrect form behavior
performance_glitch_user	secret_sauce	Simulates slow response times — tests timeout/loading behavior
error_user	secret_sauce	Generates specific UI errors during checkout
visual_user	secret_sauce	Displays visual layout bugs — suitable for visual regression testing

User Stories

- US-01 (Authentication): As a registered user, I want to log in with valid credentials so that I can access the product catalog.
- US-02 (Authentication): As a system, I want to reject invalid or locked accounts with a meaningful error message.
- US-03 (Browsing): As a shopper, I want to view all available products on the inventory page and sort them by name or price.
- US-04 (Product Detail): As a shopper, I want to view detailed information about a product before adding it to my cart.
- US-05 (Cart): As a shopper, I want to add one or more products to my cart and see the cart badge reflect the correct count.

- US-06 (Cart): As a shopper, I want to remove items from my cart either from the inventory or the cart page.
- US-07 (Checkout): As a shopper, I want to complete a purchase by providing my first name, last name, and postal code.
- US-08 (Checkout): As a system, I want to prevent checkout completion when mandatory fields are left empty.
- US-09 (Navigation): As a user, I want to access the hamburger menu to reset the application state or log out.
- US-10 (Session): As a system, I want to automatically log out the user after 10 minutes of inactivity.

3.3 Functional Requirements

ID	Module	Functional Requirement
FR-01	Authentication	System shall authenticate users with correct username and password combination.
FR-02	Authentication	System shall display 'Epic sadface: Username and password do not match...' for invalid credentials.
FR-03	Authentication	System shall display 'Epic sadface: Sorry, this user has been locked out.' for locked_out_user.
FR-04	Authentication	System shall prevent access to inventory page without a valid session.
FR-05	Inventory	System shall display all 6 products with image, name, description, and price on the inventory page.
FR-06	Inventory	System shall support sorting by: Name (A→Z), Name (Z→A), Price (Low→High), Price (High→Low).
FR-07	Product Detail	System shall display full product details when a product name or image is clicked.
FR-08	Cart	System shall allow adding products to cart; cart badge shall increment accordingly.
FR-09	Cart	System shall allow removing products; cart badge shall decrement or disappear.
FR-10	Cart	Cart contents shall persist during the same session when navigating between pages.
FR-11	Checkout Step 1	System shall accept First Name, Last Name, and Postal Code as mandatory fields.
FR-12	Checkout Step 1	System shall display a field-specific error for any missing mandatory field.
FR-13	Checkout Step 2	System shall display order summary (items, subtotal, tax, total) before confirmation.
FR-14	Checkout Confirm	System shall display 'THANK YOU FOR YOUR ORDER' confirmation upon successful checkout.
FR-15	Checkout Confirm	Cart shall be cleared and badge removed after successful order completion.

ID	Module	Functional Requirement
FR-16	Navigation	'Reset App State' menu option shall remove all cart items and reset button states.
FR-17	Navigation	'Logout' menu option shall end the session and redirect to the login page.

3.4 Non-Functional Requirements

ID	Category	Non-Functional Requirement
NFR-01	Performance	Login page shall load within 3 seconds under normal network conditions.
NFR-02	Performance	Inventory page shall render all 6 products within 4 seconds (standard_user).
NFR-03	Performance	performance_glitch_user login shall complete within 8 seconds (intentional delay).
NFR-04	Usability	All interactive elements shall be accessible via keyboard navigation.
NFR-05	Usability	Error messages shall be descriptive, user-friendly, and dismissible.
NFR-06	Security	Direct URL access to /inventory.html without a session shall redirect to login page.
NFR-07	Security	Password field shall mask input characters.
NFR-08	Compatibility	Application shall render correctly on Chrome, Firefox, and Edge (latest versions).
NFR-09	Reliability	Application shall maintain session state across page navigation within a single tab.
NFR-10	Maintainability	Automation code shall follow POM; no hard-coded locators in test classes.